SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1104

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:35**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **R. DURGA SRE / 192411189** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. DURGA SREE

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts

Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:
Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Online Banking System

The online banking system must complete transactions within 2 seconds, use multi-factor authentication, and support more than 5000 concurrent users.

Quality Assurance Analysis

1. Security Defects: Weak passwords, improper MFA implementation, session hijacking, and unauthorized access may occur.
2. Performance Defects: Transactions may take more than 2 seconds when many users access the system simultaneously.

3. Reliability Defects: Network or database failures may cause incomplete or duplicate transactions.
4. Concurrency Defects: Simultaneous fund transfers may result in incorrect account balances.
5. Transaction Defects: Incorrect transaction records may appear in transaction history.

Testing Strategies

1. Perform security testing to verify password protection, MFA, session management, and authorization.
2. Perform performance and load testing with 5000+ simultaneous users.
3. Perform stress testing by increasing the number of users beyond the expected limit.
4. Perform concurrency testing to verify that simultaneous fund transfers do not corrupt account balances.
5. Perform recovery testing by interrupting transactions and checking whether incomplete transactions are rolled back.
6. Perform audit testing to verify that every successful transaction is correctly recorded.

These techniques ensure that the banking system is secure, fast, accurate, and reliable under normal and heavy workloads.

Q2. Scenario: E-Commerce Shopping Application

System Features:
Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

E-Commerce Shopping Application

The e-commerce application should complete checkout within three steps, provide easy navigation, and work properly on both mobile and desktop devices.

Usability Challenges

1. Too many checkout pages may increase the checkout time.
2. Complicated menus may make product navigation difficult.
3. Small buttons may cause problems on mobile devices.
4. Repeatedly entering customer information can reduce efficiency.
5. Unclear error messages may confuse customers.

Usability Testing Techniques

1. Perform task-based testing by asking users to search for a product, add it to the cart, and complete checkout.
2. Measure the time taken to complete the checkout process.
3. Record the number of clicks and navigation steps required.
4. Perform responsive testing on mobile, tablet, and desktop screens.

5. Observe user mistakes and collect customer feedback.
6. Perform A/B testing to compare different interface designs.

Recommended Improvements

1. Use a three-step checkout: Cart → Delivery → Payment.
2. Provide a clear search bar and simple product categories.
3. Use large, touch-friendly buttons.
4. Provide autofill for customer information.
5. Display clear error messages.
6. Use responsive layouts for different screen sizes.

These improvements reduce navigation time and user errors, resulting in faster checkout and improved customer satisfaction.

Q3. Scenario: Hospital Management System

Modules:
Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability. **(BL5)**

Hospital Management System

The hospital management system must maintain accurate patient information, operate 24/7, and support simultaneous access by multiple departments.

Critical Test Cases

1. Verify that valid patient information is correctly registered.
2. Check that duplicate patient records are detected.
3. Verify that patient information can be updated correctly.
4. Test appointment scheduling with available and unavailable doctors.
5. Test simultaneous appointment booking by different departments.
6. Verify that authorized users can access medical records.
7. Verify that unauthorized users are denied access.
8. Check whether billing calculations are accurate.
9. Test system recovery after server or database failure.
10. Verify that multiple departments can access the system without data inconsistency.

QA Practices

1. Use data validation testing to maintain accurate patient information.
2. Use database testing to verify correct storage and retrieval of records.
3. Use security testing to prevent unauthorized access to medical records.
4. Use concurrency testing to verify simultaneous access by different departments.
5. Use availability testing to ensure continuous operation.
6. Use backup and recovery testing to prevent permanent data loss.

The constraints make data accuracy, availability, concurrency, and security the highest testing priorities. These practices help prevent data corruption, downtime, and unauthorized access.

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Smart Library Management System

The library system should display search results within 2 seconds, support users of different age groups, and require minimal learning.

Usability Evaluation

1. Perform task-based testing by asking users to search and reserve books.
2. Measure search response time and verify that it remains below 2 seconds.
3. Perform accessibility testing for font size, keyboard navigation, screen readers, and color contrast.
4. Perform first-time-user testing to determine whether users can operate the system without instructions.
5. Observe user behavior to identify confusing menus or unnecessary steps.

6. Collect feedback about user satisfaction and ease of use.

Recommended Improvements

1. Provide a large and clearly visible search bar.
2. Use simple menus and clearly labeled buttons.
3. Use readable fonts and larger controls.
4. Provide good color contrast and accessibility support.
5. Use simple search filters and suggestions.
6. Optimize database queries to provide results within 2 seconds.
7. Display clear messages when a book is unavailable.

These improvements make the system faster, easier to learn, and accessible to users of different ages and abilities.

Q5. Scenario: Airline Reservation System

Modules:
User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

Airline Reservation System

The airline reservation system must prevent duplicate bookings, support heavy holiday traffic, and provide a simple and error-free booking interface.

Major Risks

1. Two users may attempt to book the same seat simultaneously.
2. The system may become slow during peak holiday traffic.
3. Payment may succeed while ticket generation fails.
4. Incorrect seat availability may be displayed.
5. Users may enter incorrect passenger information.
6. Session timeout may interrupt the booking process.

QA Testing Techniques

1. Perform functional testing for login, flight search, seat selection, and ticket booking.
2. Perform concurrency testing to ensure that one seat cannot be booked by two users.
3. Perform load testing using large numbers of simultaneous users.
4. Perform stress testing to identify the maximum system capacity.
5. Perform security testing to protect user and payment information.
6. Perform recovery testing for payment, server, and network failures.
7. Perform database testing to maintain correct seat and booking information.

Usability Testing

1. Test the booking process with first-time users.
2. Measure the time required to complete a booking.
3. Record user errors during passenger and payment entry.
4. Test the interface on mobile, tablet, and desktop devices.
5. Use clear seat-selection indicators.
6. Provide confirmation before final booking.
7. Display simple and meaningful error messages.

Optimization

1. Use database transactions and locking to prevent duplicate seat allocation.
2. Use load balancing and scalable servers for peak traffic.
3. Use caching for frequently searched flight information.
4. Validate passenger information before payment.
5. Use rollback when payment or ticket generation fails.
6. Keep the booking process short and simple.